

STAIR CLIMBING WHEELCHAIR

A Project Report of

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degree of

Bachelor of Technology

in

Department of Electronics and Communication Engineering

By

190040360 Nandigama Apoorva

190040664 Madineni Yuvaraj Ujwal

Under the Supervision of

D.SREENIVASA RAO

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KONERU LAKSHMAIAH EDUCATION FOUNDATION,

Green Fields, Vaddeswaram- 522502, Guntur (Dist.),
Andhra Pradesh, India.

KONERU LAKSHMAIAH EDUCATION FOUNDATION

Department of Electronics and Communication Engineering



DECLARATION

The Wireless Communication Project Report entitled “**STAIR CLIMBING WHEELCHAIR**” is a record of bonafide work of **MADINENI YUVARAJ UJWAL** bearing ID No: **190040664** and **NANDIGAMA APOORVA** bearing ID No: **190040360** submitted in partial fulfilment for the award of Bachelor of Technology in **Electronics and Communication Engineering**.

I also declare that this report is of our own effort, and it has not been submitted to any other university for the award of any degree.

190040360 NANDIGAMA APOORVA

190040664 MADINENI YUVARAJ UJWAL

KONERU LAKSHMAIAH EDUCATION FOUNDATION

Department of Electronics and Communication Engineering



CERTIFICATE

The Wireless Communication Project Report entitled “**STAIR CLIMBING WHEELCHAIR**” submitted by **NANDIGAMA APOORVA** bearing ID No: **190040360** and **MADINENI YUVARAJ UJWAL** bearing ID No: **190040664** in partial fulfilment for the award of Bachelor of Technology in **Electronics and Communication Engineering**.

Name and Signature of Project Supervisor

ACKNOWLEDGMENT

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190040360 NANDIGAMA APOORVA

190040664 MADINENI YUVARAJ UJWAL

ABSTRACT

In this paper a special stair climbing robot is proposed. In today's world, robots are used for almost all purposes. Now a days many day to day problems are solved due to the implementation of robots. Many robots like balancing robot, Stair climbing robot.etc. brought revolution in the field of transporting weights. But when it comes to senior citizens or physically disabled persons it is a tough task to climb the steps.

This paper presents the structure, construction and application of an Arduino controlled special stair climbing robot attached with a wheel chair with which the person sitting in the chair directly climbs the steps. There is a special Rocker Bogie assembly of three wheels attached to the servo motors and this whole architecture is controlled with Arduino uno microcontroller board. The proposed stair climbing wheel chair bot can be controlled by a smart phone. The proposed system is very smooth and more comfortable for the person using it. The person sits on it can easily control the system with his mobile phone. This system avoids another human assistance .i.e. the one who sits on it can control it without any others help. This system is a cost effffective one.

Keywords:

Rocker Bogie, Stairs, Wheel Chair

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OBJECTIVE OF THE PROJECT:

To develop a model of stair climbing wheelchair using rocker-bogie mechanism for disabled people.

SOFTWARE TOOL:

Arduino IDE

DESCRIPTION:

1. It is a software used for programming of different Arduino Distribution Boards.
2. The Programming Language can be C or C++.
3. Some of the libraries are **Servo, Serial, Adafruit** etc.

APPARATUS USED:

- Arduino UNO
- Servo Motor
- HC-05 Module
- Arduino Bluetooth Control App

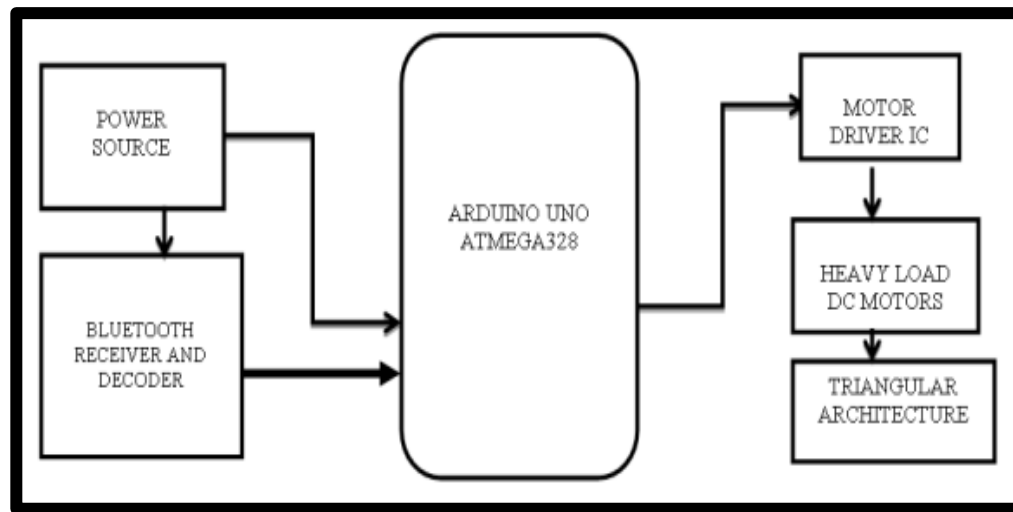
INTRODUCTION

In order to have his life on earth man must find a way to earn. As the time goes on we must remember of deprived people. We need to find a way to support such a individuals with deprived legs, hands or other parts of the body. They were too dependent on others for everything, so we need to understand their feelings and provide a way to carry out their life. It is difficult for them to migrate from one place to other. The mobility of an individual is being restricted. The small scale manufacturers were keen on their efforts to develop the cycle that suits them but it is not in the case of other transportation facilities. The tricycle invention provided them with increased mobility and independence. So we had an idea of inventing a wheel chair with rocker bogie mechanism. It consists of motor, battery, links and wheels. The suspension design used in this mechanism offers vehicle stability. The rocker bogie suspension finds its first application in Mars Exploration Rover project.

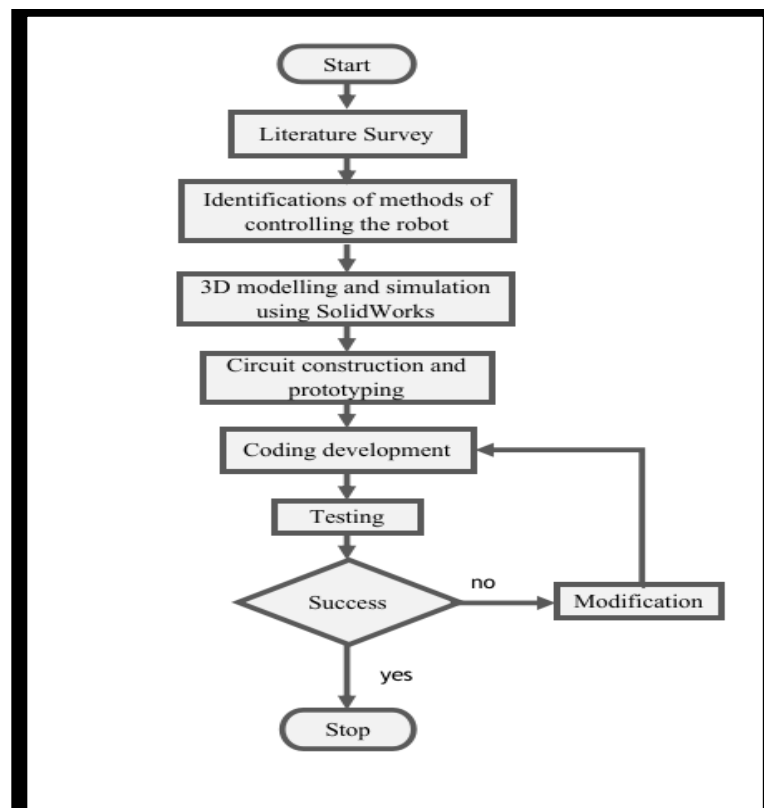
Advantages of Stair Climbing Wheelchair

1. Helps disabled people / old people in climbing the steps.
2. Wirelessly Controlled.
3. Easy to handle.

BLOCK DIAGRAM



FLOW CHART



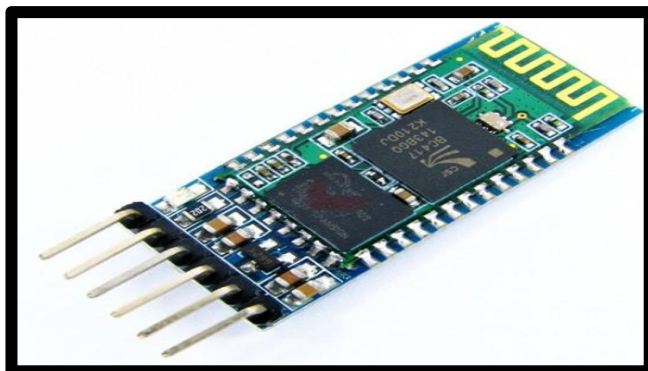
PICTURES OF APPARATUS



Arduino UNO



SG90 Servo Motor



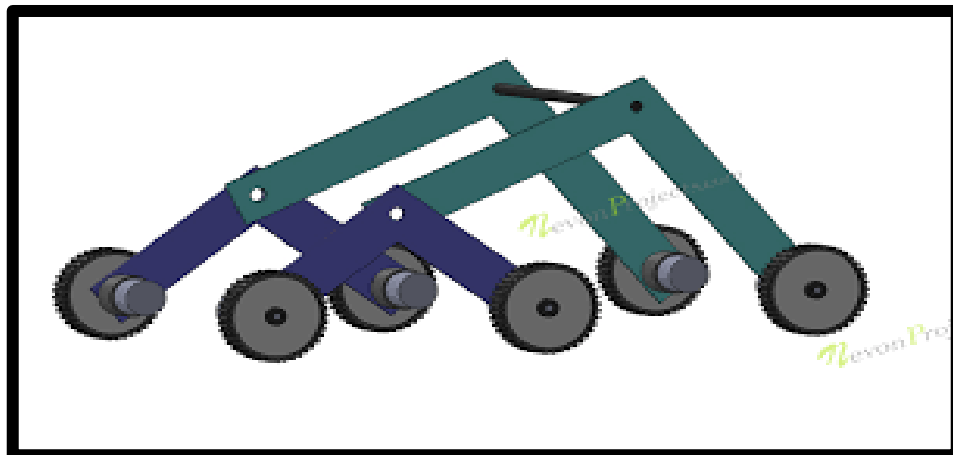
HC-05 Bluetooth Module

MECHANISM:

The mechanism used is Rocker-Bogie Mechanism.

1. The rocker-bogie system is the suspension arrangement developed in 1988 for use in NASA's Mars rover Sojourner.
2. The "rocker" part of the suspension comes from the rocking aspect of the larger, body-mounted linkage on each side of the rover.
3. The "bogie" part of the suspension refers to the smaller linkage that pivots to the rocker in the middle and which has a drive wheel at each end.

PICTURE OF MECHANISM:



PROGRAM

Arduino Code:

```
#include <Servo.h>
#include <SoftwareSerial.h>

//definicao bluetooth
#define RX 0
#define TX 1

//definicao das portas utilizadas
#define RFE 2 //roda frontal esquerda porta 0
#define RME 3 //roda meio esquerda porta 1
#define RTE 4 //roda traseira esquerda porta 2
#define RFD 5 //roda frontal direita porta 3
#define RMD 6 //roda meio direita porta 4
#define RTD 7 //roda traseira direita porta 5
#define GFE 8 //giro frontal esquerda porta 6
#define GME 9 //giro meio esquerda porta 7
#define GTE 10 //giro traseira esquerda porta 8
#define GFD 11 //giro frontal direita porta 9
#define GMD 12 //giro meio direita porta 10
#define GTD 13 //giro traseira direita porta 11

SoftwareSerial bluetooth(RX, TX); // RX, TX

//servos
Servo ServoRFD;
Servo ServoRMD;
Servo ServoRTD;
Servo ServoRFE;
Servo ServoRME;
Servo ServoRTE;
Servo ServoGFD;
Servo ServoGMD;
Servo ServoGTD;
Servo ServoGFE;
Servo ServoGME;
Servo ServoGTE;

int hor = 0; //sentido horario
int anti = 180;
int pare = 90;
int leitura = 0;
char posicao = 'n';

void setup ()
{
```

```
ServoRFD.attach(RFD);
ServoRMD.attach(RMD);
ServoRTD.attach(RTD);
ServoRFE.attach(RFE);
ServoRME.attach(RME);
ServoRTE.attach(RTE);
ServoGFD.attach(GFD);
ServoGMD.attach(GMD);
ServoGTD.attach(GTD);
ServoGFE.attach(GFE);
ServoGME.attach(GME);
ServoGTE.attach(GTE);
```

```
Serial.begin(9600); // Inicializa a serial nativa do Arduino
bluetooth.begin(9600); // Inicializa a serial via software (Onde está conectado o nosso módulo bluetooth)
```

```
    alinhado();
    parado();
}
```

```
void verificaBluetooth() { // Verifica se existe algum dado a ser lido da serial
    if (bluetooth.available()) { // verifica se existem bytes a serem lidos da porta serial virtual
        char dados = bluetooth.read(); // Lê 1 byte da porta serial
        Serial.print(dados); // Mostra esse dado lido na porta serial
```

```
        if (dados == 'p') { // Se o byte lido for igual a p
            parado();
        }
        else if (dados == 'f'){
            // Se o byte lido for igual a f
            andar();
            //delay faz o prototipo andar por por um meio segundo
            delay(500);
        }
        else if (dados == 'r'){
            // Se o byte lido for igual a r
            re();
            //delay faz o prototipo andar por por um meio segundo
            delay(500);
        }
        else if (dados == 'e'){
            // Se o byte lido for igual a e
            esquerda();
            //delay faz o prototipo andar por por um meio segundo
            delay(200);
        }
        else if (dados == 'd'){
            // Se o byte lido for igual a d
            direita();
            //delay faz o prototipo andar por por um meio segundo
```

```

        delay(200);
    }
    else if (dados == 'a'){
        // Se o byte lido for igual a a
        alinhado();
    }
}

}

void loop()
{

    verificaBluetooth();
}
void re()
{
    if(posicao == 'n'){
        ServoRFD.write(hor);
        ServoRMD.write(hor);
        ServoRTD.write(hor);
        ServoRFE.write(anti);
        ServoRME.write(anti);
        ServoRTE.write(anti);
    }
    else if(posicao == 'e' || posicao == 'd'){
        ServoRFD.write(anti);
        ServoRMD.write(anti);
        ServoRTD.write(anti);
        ServoRFE.write(anti);
        ServoRME.write(anti);
        ServoRTE.write(anti);
    }
}

void parado() {
    ServoRFD.write(pare);
    ServoRMD.write(pare);
    ServoRTD.write(pare);
    ServoRFE.write(pare);
    ServoRME.write(pare);
    ServoRTE.write(pare);
}
void andar()
{
    if(posicao == 'n'){
        ServoRFD.write(anti);
        ServoRMD.write(anti);
        ServoRTD.write(anti);
        ServoRFE.write(hor);
        ServoRME.write(hor);
        ServoRTE.write(hor);
    }
}

```

```

    }
    else if(posicao == 'e' || posicao == 'd'){
        ServoRFD.write(hor);
        ServoRMD.write(hor);
        ServoRTD.write(hor);
        ServoRFE.write(hor);
        ServoRME.write(hor);
        ServoRTE.write(hor);
    }
}
void alinhado()
{
    ServoGFD.write(90);
    ServoGMD.write(90);
    ServoGTD.write(90);
    ServoGFE.write(90);
    ServoGME.write(90);
    ServoGTE.write(90);

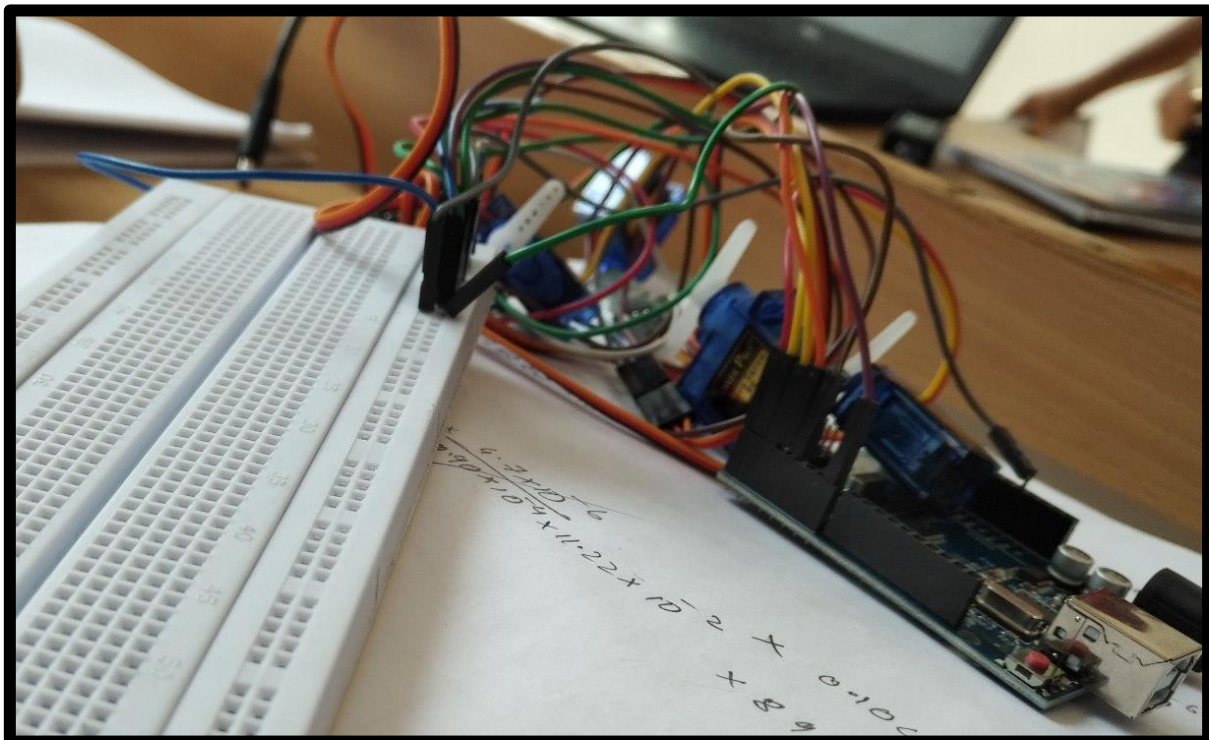
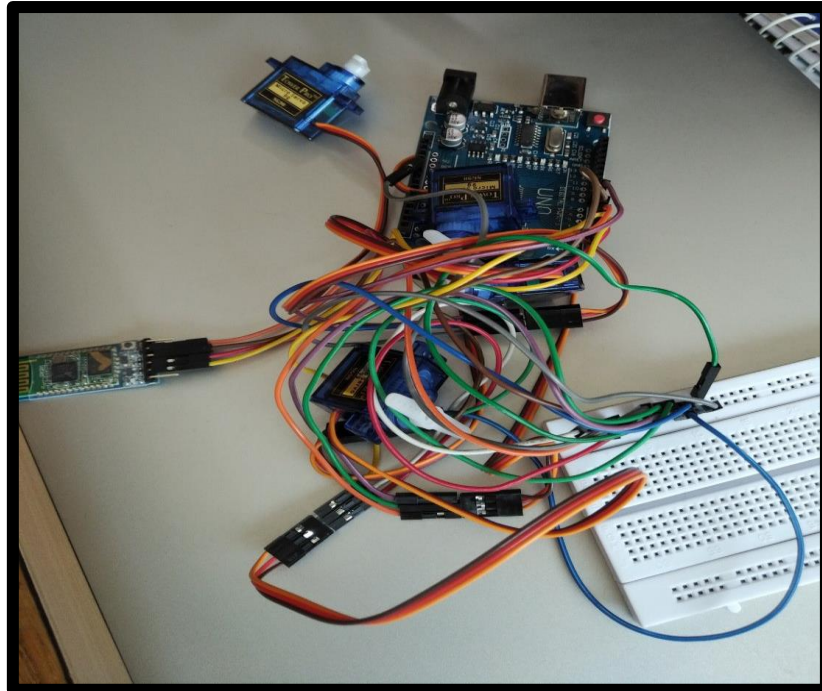
    posicao = 'n';
}
void esquerda()
{
    ServoRFD.write(hor);
    ServoRMD.write(hor);
    ServoRTD.write(hor);
    ServoRFE.write(hor);
    ServoRME.write(hor);
    ServoRTE.write(hor);

}
void direita()
{
    ServoRFD.write(anti);
    ServoRMD.write(anti);
    ServoRTD.write(anti);
    ServoRFE.write(anti);
    ServoRME.write(anti);
    ServoRTE.write(anti);
}

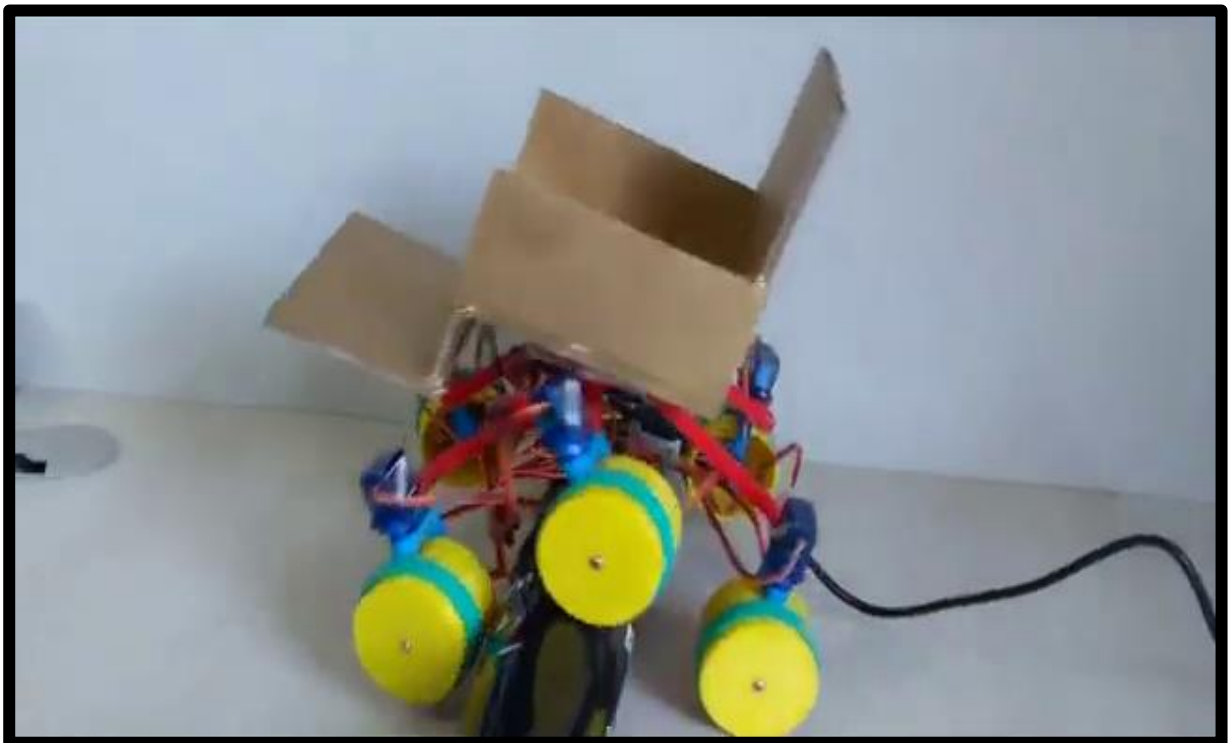
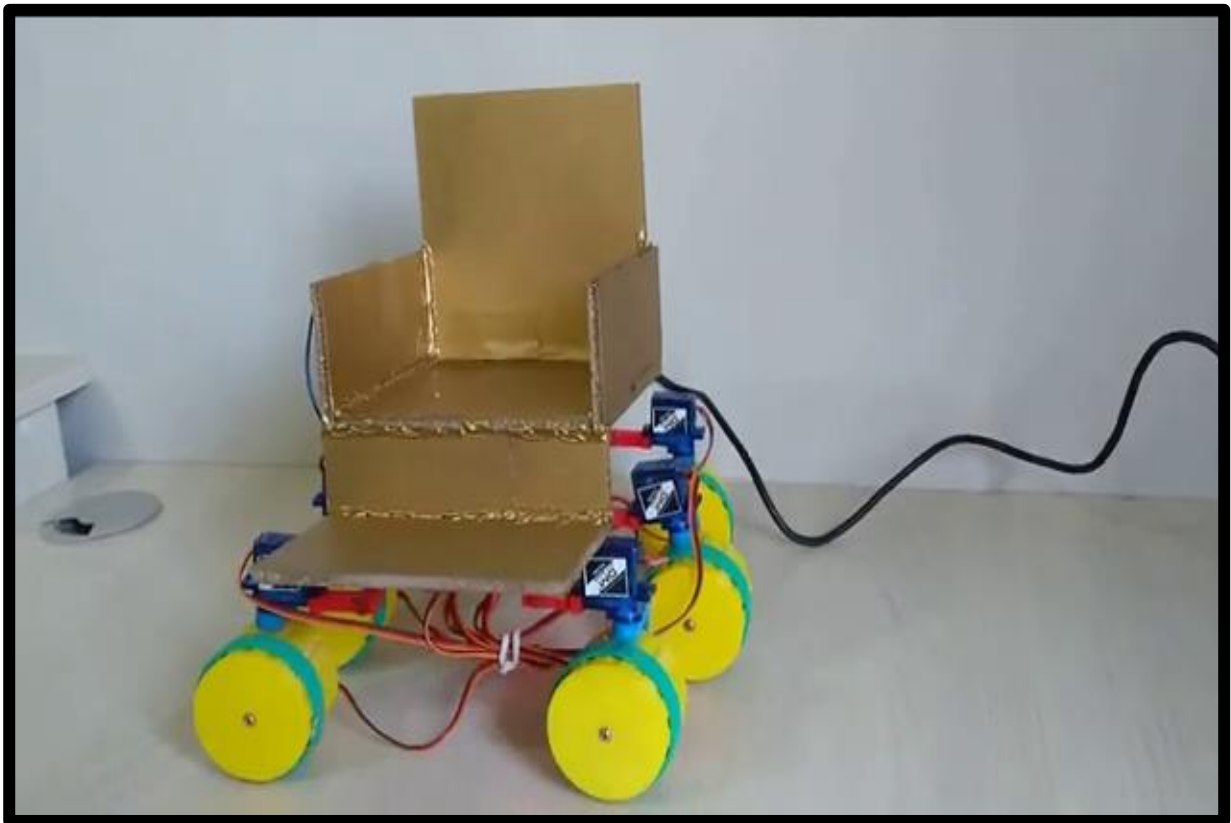
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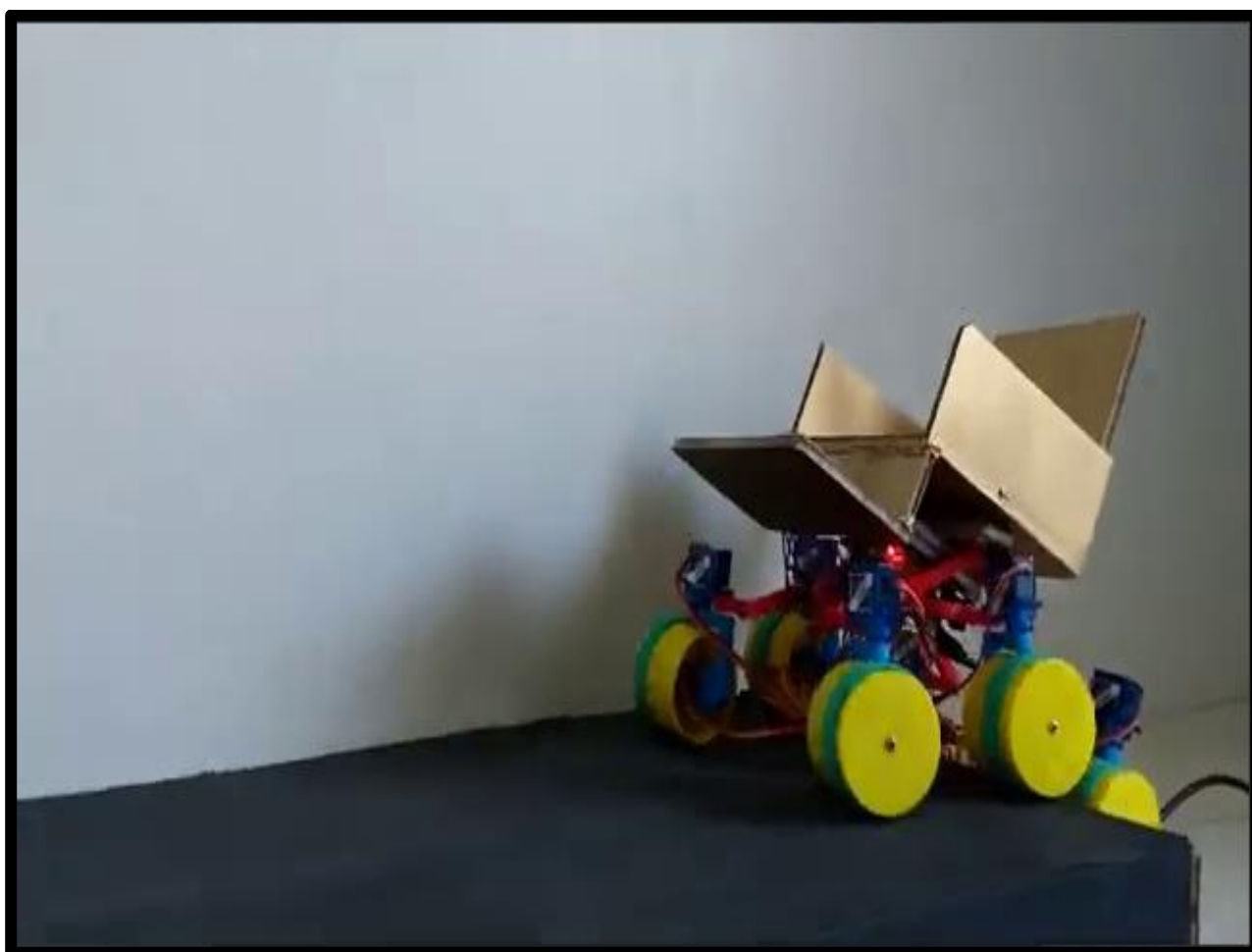
OUTPUT:

Circuit Assembly:



Prototype Assembly:





INFERENCE AND ANALYSIS

1. From the output pictures, we can depict the rocker bogie mechanism used in wheelchair to climb stairs.
2. The prototype is successful and working.

RESULT

Successfully modelled and developed the prototype of stair climbing wheelchair using rocker-bogie mechanism.

CONCLUSION

By this project, we conclude that Stair Climbing Wheelchair is useful and can be developed as a product to help disabled and old people to go up the stairs without any difficulty.

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